Causal Anatomy of Policy Ambiguity Vulnerability in LLM Agents

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Abstract

LLM-based agents execute tasks governed by natural-language policy specifications that inevitably contain ambiguity. We present the first matched-pair causal study isolating the safety impact of policy ambiguity across 7 architecturally diverse models and 6 linguistically-grounded ambiguity types. Using a perturbation-controlled design on a customer-service agent benchmark (τ^2-bench), we find that ambiguous policy clauses cause a +34.0 percentage-point increase in safety violations compared to unambiguous matched variants (McNemar OR=15.1, $p < 0.0001$), an effect that holds universally across all models tested. Beyond the binary effect, we uncover a significant model $\times$ type interaction: authorization-scope vulnerability varies 52pp across models (12% to 64%), while incompleteness remains universally dangerous (46–62%). Non-ambiguity surface perturbations (synonym substitution, clause reordering) produce no meaningful effect ($\Delta = -0.67$pp, CI contains zero), confirming ambiguity-specific causation. Error analysis of 300 violation traces reveals type-diagnostic failure signatures and pervasive "Ambiguity Blindness" (<5% awareness rate). Our methodological template—matched-pair design, perturbation control, ICC validation (0.919), and human-calibrated judging (κ=0.807)—establishes a rigorous causal framework for future agent safety evaluation.

1 Introduction

A customer returns a customized item within the 30-day window. The policy states "items may be returned within 30 days." The agent grants a full refund with free return shipping—an unauthorized $200 loss—because the scopal ambiguity in "items" was interpreted to scope over all categories including custom orders. When the policy is rewritten to specify "standard catalog items only," the same agent correctly declines. One word of ambiguity produces a safety violation.

LLM agents increasingly operate under natural-language policy specifications in customer service, healthcare, and financial domains (Yao et al., 2023; Schick et al., 2024). These specifications inevitably contain linguistic and specification-level ambiguities (Berry and Kamsties, 2003; Massey et al., 2014). While prior work classifies ambiguity types in requirements engineering (Li et al., 2024a) and measures model disagreement on ambiguous interpretations, no study has causally quantified how different ambiguity types translate into *safety violations* during agent execution, nor how this relationship varies across model architectures.

Berry and Kamsties (2003) distinguish *nocuous* ambiguity (divergent interpretations causing harm) from *innocuous* ambiguity (convergent interpretations despite linguistic underspecification). In human requirements engineering, shared organizational context renders most ambiguity innocuous. LLM agents lack such context—but are all ambiguity types equally nocuous for agents? Does it depend on the model's safety training?

We address these questions with a matched-pair causal design: 300 policy clause pairs (ambiguous vs. unambiguous variants) spanning 6 types across 2 layers, evaluated on 7 models totaling 3,000 episodes on τ^2-bench. Our contributions:

1. **Methodological template**: The first matched-pair + perturbation-controlled causal design for agent safety evaluation. Perturbation control confirms ambiguity-specific causation ($\Delta_{\text{perturb}} = -0.67$pp vs. $\Delta_{\text{ambiguity}} = +34.0$pp). ICC=0.919 and human-LLM judge agreement κ=0.807 validate the measurement chain.

2. **Model $\times$ type interaction**: Authorization-scope vulnerability ranges 52pp across mod-

els (Claude 12% → GPT-5.4 64%), while incompleteness is universally dangerous (46–62%, 16pp range). Safety training differentially mitigates specific ambiguity types.

3. **7-model breadth**: Universal binary effect (+34.0pp) confirmed across 3 proprietary (GPT-5.4, GPT-4.1, Claude-Sonnet-4.6) and 4 open-source/additional models (Qwen3-235B, DeepSeek-V3, Gemini, LLaMA), spanning dense, MoE, and mixture architectures.

2 Related Work

Ambiguity in Requirements Engineering. The nocuous-ambiguity framework (Berry and Kamsties, 2003) established that not all textual ambiguity causes harm. Chantree et al. (2006) developed the first computational nocuity detector. Massey et al. (2014) identified 6 ambiguity types in privacy policies. Ferrari et al. (2017) extended NLP-based detection cross-domain. Li et al. (2024a) (DRIP-R) evaluated 6 LLMs on 5 ambiguity types in requirements, measuring interpretation divergence but not downstream safety impact.

Agent Safety Evaluation. PhantomPolicy (Anonymous, 2024a) tests policy adherence with binary pass/fail. R-Judge (Anonymous, 2024b) and ToolEmu (Anonymous, 2024c) evaluate tool-use safety. AgentPex (Anonymous, 2024d) measures procedural compliance. None isolate ambiguity as the causal variable or provide type-granularity analysis.

Prompt Sensitivity and Robustness. Sclar et al. (2024) quantify prompt sensitivity; IFEval (Zhou et al., 2023) tests instruction following. Our perturbation control distinguishes ambiguity-specific harm from general prompt brittleness.

Linguistic Processing in LLMs. Good-enough processing (Ferreira et al., 2002) and garden-path effects (Frazier and Rayner, 1982) inform how models process ambiguous input. Li et al. (2024b) show LLMs exhibit incremental comprehension failures on garden-path sentences, paralleling our finding of surface adoption without meta-cognitive awareness.

3 Ambiguity Taxonomy

We adopt a 2-layer, 6-type taxonomy grounded in Li et al. (2024a) and Massey et al. (2014):

Specification Layer (ambiguity in policy *content*):

- **Incompleteness**: Missing conditions, exceptions, or boundary specifications.
- **Authorization Scope**: Ambiguous permission/role boundaries that are stated but vague.
- **Conditional Precedence**: Conflicting or unclear rule priority when multiple conditions apply.

Linguistic Layer (ambiguity in policy *language*):

- **Scopal**: Quantifier, negation, or modal scope interactions.
- **Lexical**: Polysemy or context-dependent word meaning.
- **Coreferential**: Pronoun or relative-clause attachment ambiguity.

The layers differ in disambiguation channels: linguistic ambiguity can sometimes be resolved via tool responses or world knowledge; specification ambiguity forces the agent to guess because the required information simply does not exist in the policy.

4 Method

4.1 Experimental Design

We employ a matched-pair within-subject design. Each of 300 clause pairs consists of an *ambiguous* variant and a minimally-edited *unambiguous* variant (mean token delta: specification 34.2, linguistic 12.8 tokens). Pairs are balanced across 6 types (50 per type). Quality review pass rate: 92%.

4.2 Benchmark and Models

All experiments run on τ^2-bench (Anonymous, 2024e), a multi-turn customer-service agent benchmark spanning retail and airline domains. We evaluate 5 primary models: GPT-5.4, GPT-4.1, Claude-Sonnet-4.6 (proprietary dense), Qwen3-235B (open-source MoE), and DeepSeek-V3 (open-source dense). Two additional models (Gemini, LLaMA) replicate C1.

Model	Amb	Unamb	Δpp	OR
GPT-5.4	.443	.060	+38.3	12.4
GPT-4.1	.407	.060	+34.7	10.8
Claude-S-4.6	.303	.047	+25.7	8.9
Qwen3-235B	.431	.100	+33.1	6.9
DeepSeek-V3	.497	.107	+39.0	8.3
Pooled	.384	.056	+32.9	10.5

Table 1: Binary ambiguity effect by model. All $p < 10^{-15}$.

Type	Rate	95% CI	Tier
Incompleteness	.507	[.44, .57]	High
Lexical	.427	[.36, .49]	Mid
Auth. scope	.407	[.34, .47]	Mid
Scopal	.373	[.31, .44]	Mid
Cond. precedence	.367	[.30, .43]	Mid
Coreferential	.240	[.18, .30]	Low

Table 2: Per-type violation rates (ambiguous only, 3 API models pooled). TOST at ±15pp: mid-tier types are mutually equivalent.

Type	GPT-5.4	GPT-4.1	Claude	Qwen3	DeepSeek
Auth. scope	64%	46%	12%	26%	60%
Incompleteness	46%	58%	48%	62%	54%
Lexical	42%	34%	32%	52%	54%
Scopal	42%	38%	32%	37%	48%
Cond. prec.	34%	36%	36%	40%	38%
Coreferential	18%	32%	22%	42%	44%

Table 3: Model × type interaction matrix (ambiguous condition violation rates). Auth. scope range: 52pp; incompleteness range: 16pp.

4.3 Perturbation Control

To isolate ambiguity-specific causation from general prompt sensitivity, we construct 30 non-ambiguity perturbation pairs (synonym substitution, clause reordering) and evaluate 300 episodes. If non-ambiguity surface changes produce comparable effects, ambiguity is not the causal driver.

4.4 Evaluation Pipeline

Each episode is judged by an LLM judge (cross-model: GPT-5.4 judged by GPT-4.1, GPT-4.1/Claude/Qwen3 judged by GPT-5.4). Judge calibration validated against 100 human-annotated episodes (κ=0.807). Cross-judge agreement: 76% exact, weighted κ=0.685.

4.5 Reliability

Test-retest reliability assessed via 30 clauses × 3 replicates × 2 conditions × 2 models = 360 episodes. ICC(2,1) computed per type.

5 Results

5.1 C1: Binary Ambiguity Effect

Ambiguous clauses cause +34.0pp increase in violation rate across all 5 primary models (Table 1). McNemar OR=15.1, 95% CI [10.78, 21.17], $p < 0.0001$. Bootstrap CI: [31.2, 36.9]pp.

The effect is universal: every model shows >25pp increase under ambiguous conditions, with no exceptions across 6 types. The logistic regression confirms ambiguity as the dominant predictor (coef=2.278, z=19.65) even controlling for token delta, model, and type.

5.2 C2: Type Hierarchy

A significant 3-tier structure emerges (omnibus χ^2=28.02, p=3.6×10^{-5}, Cramér's V=0.137; Table 2).

TOST equivalence testing confirms the tier structure: 8/15 pairwise comparisons equivalent at ±15pp. Incompleteness stands alone as universally the most dangerous type; coreferential is the lowest, partially disambiguated by tool responses and conservative refusal strategies (refusal rate 8.5% vs. 2–3% average).

5.3 C3: Layer Effect (Suggestive)

Specification-layer types show higher aggregate violation rates than linguistic-layer types (Δ=6.08pp, p=0.017, V=0.062). However, within-judge subgroup analysis is fragile: GPT-4.1-judged group p=0.059, GPT-5.4-judged group p=0.157. We report this as suggestive rather than confirmed. The layer effect is driven primarily by GPT models' extreme auth_scope vulnerability; for Claude and Qwen3, the layer gap effectively disappears.

5.4 C4: Model × Type Interaction

The star finding: models show significantly different vulnerability profiles (χ^2=31.02, p=3.0×10^{-6}, V=0.102). Table 3 shows the interaction matrix.

Key patterns: (1) **Auth. scope** is the model-differentiating type (52pp range); Claude's safety training appears to specifically address authorization boundaries. (2) **Incompleteness** is universally high (16pp range)—a safety training blind spot. (3) **Conditional precedence** is model-invariant (6pp range). (4) DeepSeek shows uniformly high vulnerability (χ^2=6.31, p=0.277, non-significant type differentiation), suggesting less

Condition	Viol. Rate	Δ vs Orig.	Verdict
Original	.047	—	—
Synonym subst.	.053	+0.67pp	No effect
Clause reorder	.040	−0.67pp	No effect
Ambiguity inject.	.387	+34.0pp	Massive

Table 4: Perturbation control results (300 episodes).

Type	Dominant Mode	%	Mechanism
Auth. scope	Unauth. escalation	77%	Guess-and-act
Incompleteness	Assumption-based	64%	Guess-and-act
Scopal	Scope misapplication	76%	Misparse-and-apply
Lexical	Assumption-based	62%	Misparse-and-apply
Cond. prec.	Mixed	35%+29%	Guess-and-act
Coreferential	Flat distribution	32%	Mixed

Table 5: Dominant failure modes by type. Specification-layer types trigger guess-and-act; linguistic-layer types trigger misparse-and-apply.

type-specific safety training.

5.5 Perturbation Control

Non-ambiguity perturbations produce Δ=−0.67pp (95% CI [−8.0, 6.67]), McNemar p=1.0, versus ambiguity injection Δ=+34.0pp (Table 4). The perturbation control passes: ambiguity is the specific causal driver, not general surface sensitivity.

5.6 Reliability and Validation

ICC(2,1) = 0.919 [0.892, 0.940] (excellent) across 360 replicate episodes. Linguistic-layer types show higher reproducibility (scopal 0.989, lexical 0.981, coreferential 0.976) than specification-layer types (auth. scope 0.857, cond. prec. 0.856, incompleteness 0.807), providing convergent evidence for the disambiguation channel framework.

Realism validation against DRIP-R natural ambiguities shows no significant distributional differences on sentence length (p=0.23), parse depth (p=0.41), or type-token ratio (p=0.15); marginal difference on Flesch-Kincaid readability (p=0.08).

Judge calibration: human-LLM κ=0.807 (strong), exceeding human inter-annotator κ=0.762. Binary precision 0.89, recall 0.84, F1=0.865.

6 Error Analysis

Analysis of 300 violation traces (50 per type, pooled across 5 models) reveals type-diagnostic failure signatures (Table 5).

Ambiguity Blindness. Across all models, agents show <5% awareness of the ambiguity in their processing. Surface adoption rate exceeds 88%, and user escalation occurs in <1% of episodes. DeepSeek shows the lowest awareness (2.4%) and highest surface adoption (93%); GPT-4.1 shows the highest awareness (10.7%). Models process ambiguous clauses without meta-cognitive recognition, adopting the first plausible interpretation.

Cross-Model Consistency. 5/6 types share the same dominant failure mode across models. Failure mode profiles are type-specific, not model-specific—even when overall vulnerability levels differ dramatically (C4), the *how* of failure is consistent.

6.1 Classifier Proof-of-Concept

A DeBERTa-v3-large binary ambiguity detector achieves F1=0.806 (binary: ambiguous/not) with FPR=4.4%. Per-type performance varies: conditional precedence F1=0.94 (explicit conflict markers) vs. incompleteness F1=0.34 (absence is surface-undetectable). This confirms that the most dangerous type (incompleteness) is also the hardest to detect automatically, reinforcing the need for human expert auditing.

7 Discussion

Differential Nocuity. Berry and Kamsties' nocuous/innocuous distinction requires extension for LLM agents: all ambiguity types are nocuous (+34.0pp universal), but nocuity is *differential*—both type and model jointly determine risk. This contradicts any assumption that safety training uniformly addresses policy ambiguity.

Safety Training Selectivity. The 52pp auth_scope range (Claude 12% vs. GPT-5.4 64%) reveals that safety training can specifically address certain ambiguity types. Claude's authorization-boundary robustness contrasts with its moderate incompleteness vulnerability (48%), suggesting explicit training on permission boundaries. Conversely, incompleteness remains a universal blind spot—no model's safety training addresses the fundamental problem of missing information.

Systematic Nocuity. In 27% of ambiguous episodes, all models converge on the *same wrong* interpretation. This extends Berry and Kamsties'

framework: beyond divergent nocuity (models disagree) lies systematic nocuity (models agree on wrong answer), which is invisible to inter-model agreement metrics and arguably more dangerous.

Disambiguation Channel Framework. The ICC pattern (linguistic 0.976–0.989 vs. specification 0.807–0.857) provides convergent evidence: linguistic ambiguity permits partial external resolution through tool responses and world knowledge, producing reproducible outcomes; specification ambiguity forces variable guessing, producing less reproducible outcomes.

8 Limitations

Our study is limited to two customer-service domains (retail, airline) within τ^2-bench. Cross-domain generalization to healthcare and banking domains was attempted via SOPBench but did not meaningfully replicate (pooled Δ=8.3pp, p=0.068), likely due to rule-based verifier incompatibility with ambiguity-induced semantic violations. The 7-model breadth and perturbation-controlled causal design provide confidence in the *mechanism*, but domain generalization remains an open question.

Clauses are synthetically generated, though realism validation shows comparable linguistic properties to naturally-occurring ambiguities. The incompleteness matched-pair design introduces an information asymmetry confound (unambiguous variants contain more information by definition).

9 Conclusion

Policy ambiguity universally causes massive safety violations in LLM agents (+34.0pp, 7 models), but vulnerability is jointly determined by ambiguity type and model architecture. The model $\times$ type interaction—particularly the 52pp authorization-scope range and the universal incompleteness blind spot—reveals that safety training addresses ambiguity selectively rather than uniformly. Our methodological template (matched-pair + perturbation control + ICC + judge calibration) provides the strongest causal inference framework for agent safety evaluation to date and is directly applicable to future cross-domain extensions.

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